(EXPLAINABLE) FRAUD DETECTION USING DIFFUSION TRAJECTORIES

FINAL PROJECT REPORT

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1 Problem Statement & Motivation

Fraud detection systems are commonly designed to assign scalar anomaly scores to transactions, enabling automated identification of potentially fraudulent activity. While effective in practice, these approaches often provide limited interpretability, offering little insight into why a transaction is flagged as anomalous. This lack of transparency can hinder trust, complicate auditing processes, and make it difficult for analysts to act on model outputs in high-stakes financial environments [3] [5]. In many real-world deployments, the ability to justify and explain predictions is as important as the predictions themselves. In addition, many traditional fraud detection approaches rely on supervised learning techniques that require labeled fraudulent data. However, fraud is inherently rare, continuously evolving, and often underrepresented in available datasets, making it difficult to obtain comprehensive and up-to-date labels. Consequently, supervised models may struggle to generalize to emerging fraud patterns or previously unseen attack strategies [2].This limitation has motivated the use of unsupervised and one-class learning methods that model normal behavior and identify deviations from it [4].

In this project, we explore an unsupervised generative modeling approach based on diffusion models trained exclusively on legitimate transactions. Diffusion models have recently demonstrated strong performance in learning complex data distributions through an iterative denoising process, gradually transforming noise into samples that follow the training distribution [1] [6]. By learning the manifold of normal transactions, deviations from this learned distribution could be used as indicators of anomalous behavior without requiring labeled fraud examples. Beyond standard reconstruction-based signals, this work is motivated by the hypothesis that the internal dynamics of the diffusion process - specifically, the denoising trajectories which may encode additional structure that could support interpretability. In particular, the sequence of intermediate states generated during reverse diffusion may reflect how a transaction evolves toward the learned data manifold, potentially revealing differences between normal and anomalous samples. Prior work on score-based generative modeling suggests that such trajectories are closely tied to the geometry of the learned data distribution [6]. The motivation behind this approach is twofold. First, we aim to improve fraud detection in settings where labeled data is scarce or unreliable by leveraging only normal transaction data. Second, we seek to investigate whether diffusion-based trajectory representations can provide interpretable signals that explain why a transaction deviates from normal behavior. Rather than assuming that trajectory geometry will necessarily yield improved explainability, this project empirically evaluates whether such trajectory-based features offer meaningful advantages over standard reconstruction-based anomaly signals.

2 Methodology

2.1 Dataset

We used the Credit Card Fraud Detection dataset from Kaggle linked here, which consists of anonymized transaction records derived from real credit card transactions. The dataset contains numerical features (resulting from a PCA transformation of the original attributes) along with a highly imbalanced binary label indicating fraudulent and legitimate transactions.

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2.2 Overview

Our approach was divided into two main parts. In Part I, we evaluate the effectiveness of diffusion models for tabular fraud detection using reconstruction-based anomaly signals. In Part II, we extend the analysis by examining whether the geometry of denoising trajectories can provide additional discriminatory or explanatory information. Finally, we perform ablation and visualization studies to assess the contribution of different components and validate findings qualitatively.

2.3 Part I: Diffusion for Tabular Fraud Detection

2.3.1 Data Preprocessing

The Credit Card Fraud Detection dataset is first subjected to standard preprocessing, including data cleaning and feature-wise normalization. All input features are scaled to ensure numerical stability and to prevent features with larger magnitudes from disproportionately influencing the learning process. The model is trained exclusively on legitimate transactions, thereby formulating the problem as a one-class learning task.

2.3.2 Forward Diffusion Process

Let x_0 denote a legitimate transaction sample. The forward diffusion process progressively corrupts x_0 by adding Gaussian noise according to a predefined variance schedule over a sequence of timesteps $t \in \{1, \ldots, T\}$. This yields a sequence of latent variables $x_1, x_2, \ldots, x_T$, where the final state x_T approaches an isotropic Gaussian distribution. Each timestep corresponds to a progressively noisier representation of the original input.

2.3.3 Timestep Conditioning

To enable the model to operate effectively across varying noise levels, timestep information is incorporated as a conditioning signal. A learned timestep embedding encodes the diffusion step and is provided alongside the noisy input x_t. This conditioning allows the model to adapt its predictions based on the level of corruption present in the input, effectively guiding the denoising process.

2.3.4 Noise Prediction with a Residual MLP

The denoising model is parameterized using a residual multi-layer perceptron (MLP) that takes as input the noisy sample x_t concatenated with its timestep embedding. The network is trained to predict the noise component ϵ added during the forward diffusion process. Residual (skip) connections are employed to facilitate optimization and improve gradient flow. This architecture enables the model to focus on learning the residual mapping between the noisy input and the underlying signal, rather than reconstructing the input from scratch.

2.3.5 Training Objective

The model is optimized using a mean squared error loss between the predicted noise $\hat{\epsilon}$ and the true noise ϵ. The objective is defined as:

$$\mathcal{L} = \mathbb{E}_{x_0,\epsilon,t} \left[\|\epsilon - \hat{\epsilon}(x_t, t)\|^2 \right], \tag{1}$$

where x_t denotes the noisy version of x_0 at timestep t, and the expectation is taken over the data distribution, noise samples, and timesteps.

2.3.6 Anomaly Scoring

At inference time, anomaly scores are derived from the model's denoising performance on a given transaction. Specifically, the discrepancy between the predicted noise and the actual noise is used as a proxy for how well the sample conforms to the learned distribution of legitimate transactions.

Samples that lie close to the data manifold are expected to be denoised accurately, resulting in low prediction error. In contrast, anomalous transactions tend to deviate from the learned distribution, leading to higher reconstruction or denoising errors. These errors are aggregated across timesteps to produce a final anomaly score, where larger values indicate a higher likelihood of fraud.

2.4 Part II: Trajectory Geometry and Explainability

2.4.1 Trajectory Extraction

For this part of the project, the full reverse denoising process is run for 500 test transactions. These samples contained all the 67 fraud cases and 433 randomly selected legitimate examples, ensuring a stratified sample. The state of each transaction is recorded every 5 de-noising steps, producing a trajectory of shape [500 samples x 21 steps x 30 features]

2.4.2 Geometric Analysis

Next, we computed five geometric features per trajectory to characterize the shape of each transaction's path through feature space during the denoising process. These features were:

- **Total Path length:** the Sum of the L2 distances between consecutive states
- **Mean Curvature:** Average angle between consecutive displacement vectors
- **Directional consistency:** Average cosine similarity between consecutive displacements
- **Straightness:** Ratio of net displacement to total path length
- **Step length variance:** the variance in the size of individual denoising steps

2.4.3 Augmented Anomaly Score

Next, to formally test whether geometric features (that we calculated in the previous section) could augment the baseline reconstruction error, we establishes a new combined score:

$$\text{Score}_{\text{combined}} = w_{\text{base}} \times \text{score}_{\text{baseline}} + w_{\text{geo}} \times \text{score}_{\text{geometry}}$$

The geometric score combines normalized Path length, curvature, step length variance, and inverted directional consistency.

2.4.4 Interpretability Analysis

As discussed in previous sections, it is extremely important to have a component of human interpretibility to a project like fraud detection. So, to understand where within the reconstruction process the fraud signal is the strongest, the reconstruction errors were evaluated at 20 evenly spaced timestep bins across the noise schedule, for all 30 features, separately. Additionally, it is extremely important to note that this analysis is separate from the geometric analysis, and the calculations done in the geometric analysis are not used to drive the interpretability.

3 Results

In this section, we present the experimental results of our approach, beginning with the findings from Part I (diffusion-based fraud detection) and followed by the analysis in Part II (denoising trajectory geometry and related studies).

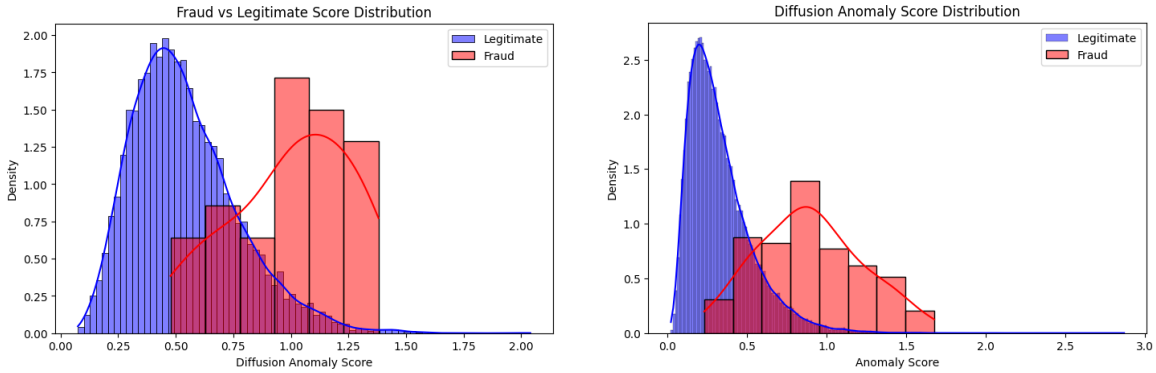
3.1 Evaluating Diffusion for Tabular Fraud Detection

We evaluate the effectiveness of diffusion models for tabular fraud detection by comparing two model variants. Performance is measured using AUROC and Average Precision to account for the highly imbalanced class distribution, where AUROC captures ranking ability and Average Precision reflects performance on the minority (fraudulent) class. The baseline model (Version A) is a standard MLP trained on noisy legitimate transactions with $T = 1000$ diffusion steps over 60 epochs. It achieves an AUROC of 87%, but exhibits limited ability to detect fraudulent transactions, as indicated by a low Average Precision of approximately 5%. The improved model (Version B) incorporates residual connections and timestep embeddings (described in Section 2)and is trained with fewer diffusion steps ($T = 100$). These enhancements allow the model to better capture the denoising process, resulting in significant performance gains, with AUROC increasing to 95% and Average Precision rising to 27%. The quantitative performance of both model variants is summarized in Table 1.

Model	AUROC	Avg Precision	T	Epochs	Notes
Ver A	0.8723	0.0551	1000	60	Plain MLP
Ver B	0.9509	0.2711	100	60	Residual MLP + Timestep Embeddings

Table 1: Performance comparison between Version A and Version B.

To gain further insights into model behavior, we also analyzed the distribution of anomaly scores as a function of diffusion density for both variants. As seen in Figure 1, Version B showed improved separation between legitimate and fraudulent transactions, although some overlap remains.



(a) Version A (b) Version B

Figure 1: Anomaly score vs diffusion density for Version A (left) and Version B (right)

3.2 Geometric Analysis Results

The geometric features were virtually identical for the fraudulent and legitimate classes:

Feature	Legitimate	Fraud
Total Path Length	22.734514	22.781794
Mean Curvature	1.571648	1.573259
Max Curvature	1.925543	1.915787
Mean Direction Consistency	-0.000908	-0.002439
Step Length Variance	0.181156	0.184834
Net Displacement	4.744353	4.738624
Straightness	0.208538	0.208142

Table 2: Comparison of geometric features between legitimate and fraudulent trajectories

Furthermore, the augmented score grid search confirmed this finding, selecting optimal weights of baseline =1.0 and geometry =0.0. This reinforces the idea that the trajectory geometry carries no discriminative signals for fraud detection in this setting. This is because the shape of the denoising oath is determined mainly by the model's learned reverse process instead of the characteristics of the input transactions.

3.3 Interpretability Results & Explanation

We first looked at the **Most Discriminative timesteps**. We found that denoising steps 41, 20, and 46 (out of T=100) showed the largest error gap between fraud and legitimate cases. Fraud is most differentiable at mid noise levels.

Next, we tried to find features that resulted in a higher legitimate vs fraud error gaps

Feature	Error Gap
V_{17}	0.9663
V_{16}	0.8229
V_1	0.7792
`amount`	0.7593
V_2	0.7591
V_{12}	0.7344
V_5	0.7186
V_{20}	0.7090
V_3	0.7068
V_4	0.6994

Table 3: Top anomalous features ranked by error gap

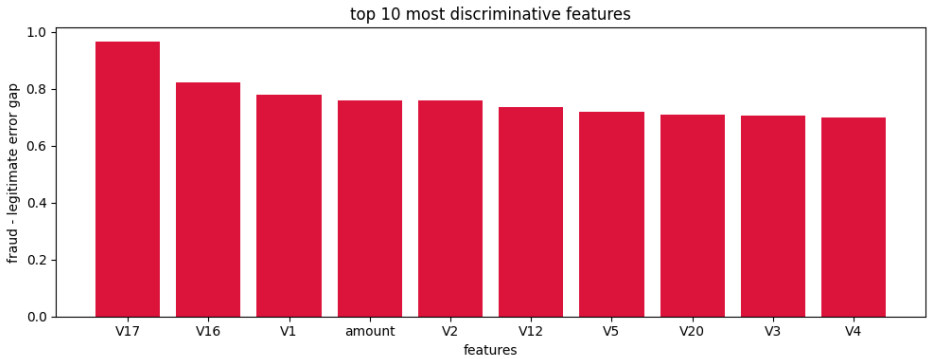


Figure 2: Top 10 anomalous features

This led us to build an explanation generator. Instead of just outputting a number, the system can now say: this transaction was flagged because these specific features deviated from normal at these specific reconstruction steps. That is the difference between a black box and something a fraud analyst can actually act on. This is what an explanation output looks like:

Transaction #0 Analysis	
True Label	FRAUD
Decision	Flagged as fraudulent
Anomaly Score	0.2702
Threshold (80th pct)	0.2371
Denoising Analysis	
Peak Reconstruction Difficulty	Steps 0–99, error = 1.68809
Flagging Reason	V_{16}, time, V_{20} deviated jointly
Feature Contributions	V_{16} (25.6079), time (6.6299), V_{20} (2.7149)
Trajectory Geometry	
Total Path Length	21.6387
Mean Curvature	1.5267 rad
Trajectory Straightness	0.2606

Table 4: Detailed anomaly analysis for Transaction #0

3.4 Visualization Results

Lastly, to validate everything visually, we built three different types of visuals:

3.4.1 PCA

First, we compressed the 25D trajectory representation down to 2D using PCA. The two principal components explained 67.3% of the total variance across the 500 sample trajectory representations. PC1 alone captures 58.5% variance and produces a drastic separation between the fraudulent and legitimate transactions. The legitimate transactions cluster tightly on the left while the fraudulent transactions shift towards the left.

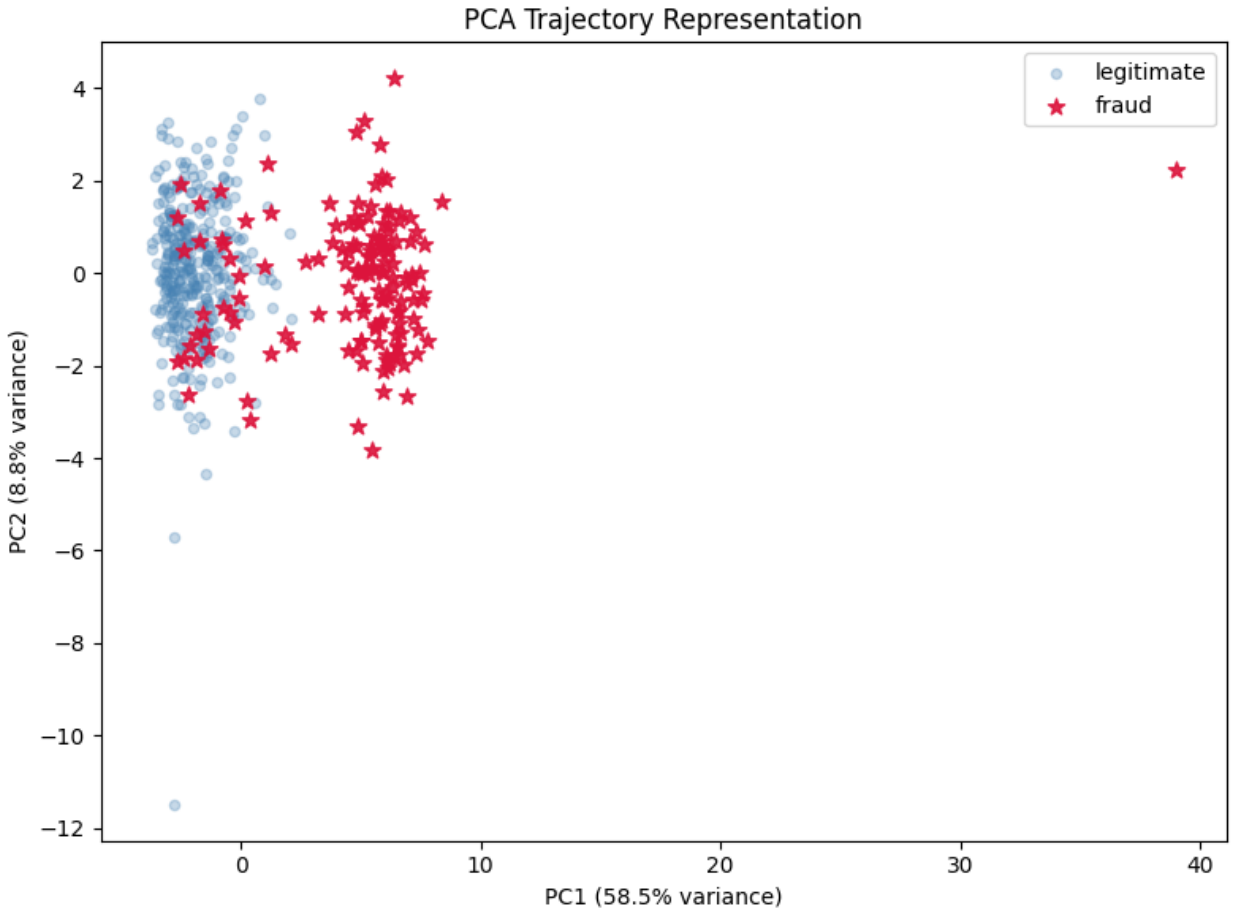


Figure 3: PCA Trajectory Representation

3.4.2 T-SNE

The T-SNE plots are a bit more revealing. It shows two distinct islands. We can see a large diffused cloud on the left containing (mostly) the legitimate cases, and on the right, there is a tighter cluster containing the fraudulent cases. A tighter cluster is a prominent finding, since it proves that not only are the fraud and legitimate cases different from each other, but that the fraudulent cases are similar to one another. This is proof that the model has learned a coherent representation of what makes a transaction anomalous rather than simply responding to random noise.

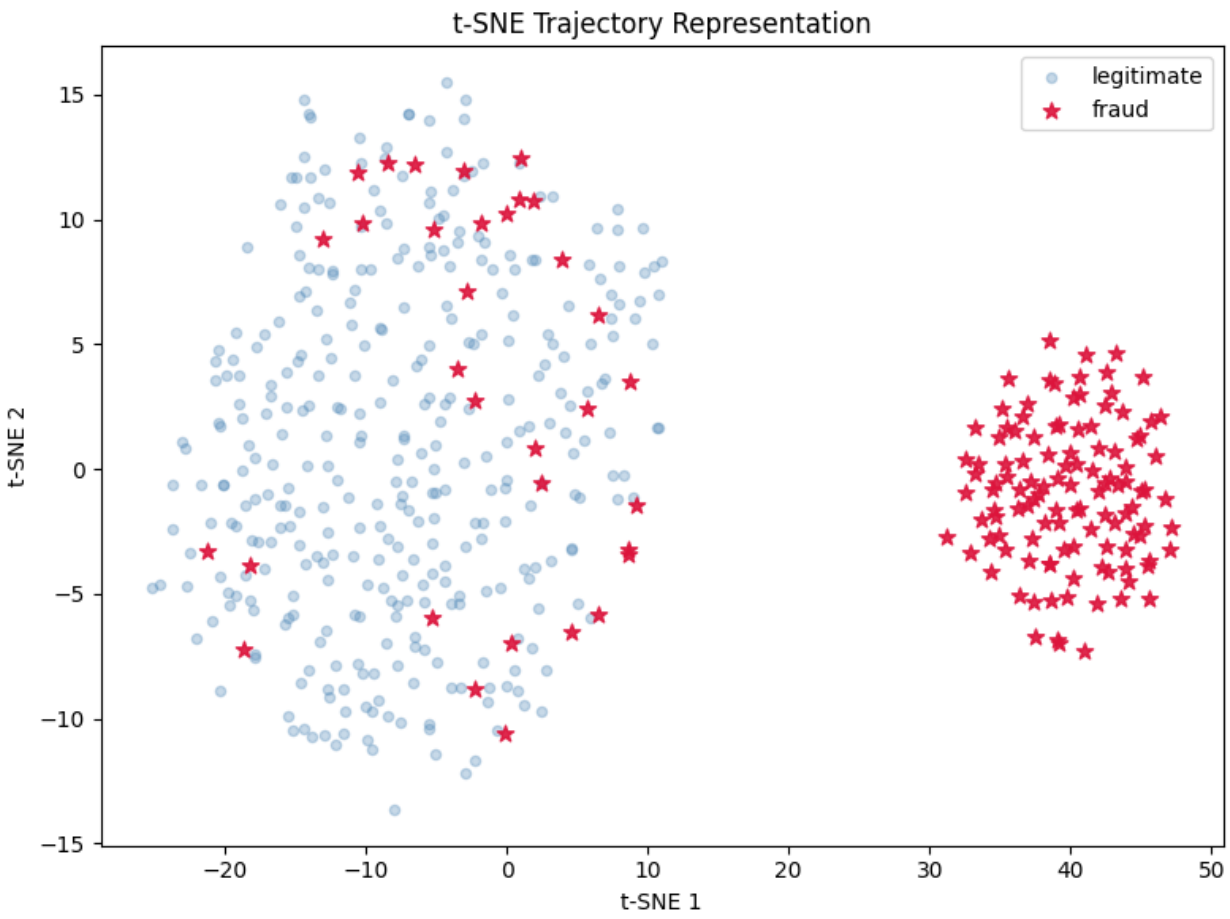


Figure 4: T-SNE Trajectory representation

3.4.3 Individual Trajectory Visualization

We randomly visualized a few fraudulent and legitimate pairs by projecting their 21 snapshots from 30D feature space to 2D using PCA fitted on each pair combination. A consistent pattern appeared across almost all pairs. The fraudulent trajectories formed tangles looping paths (unlike legitimate transactions), suggesting that the model had trouble reconstructing the endpoints. Even though there is a visual difference between how these paths look, it is important to note that these are random pairs that are compared, and that the geometric features that are calculated are actually similar to each other.

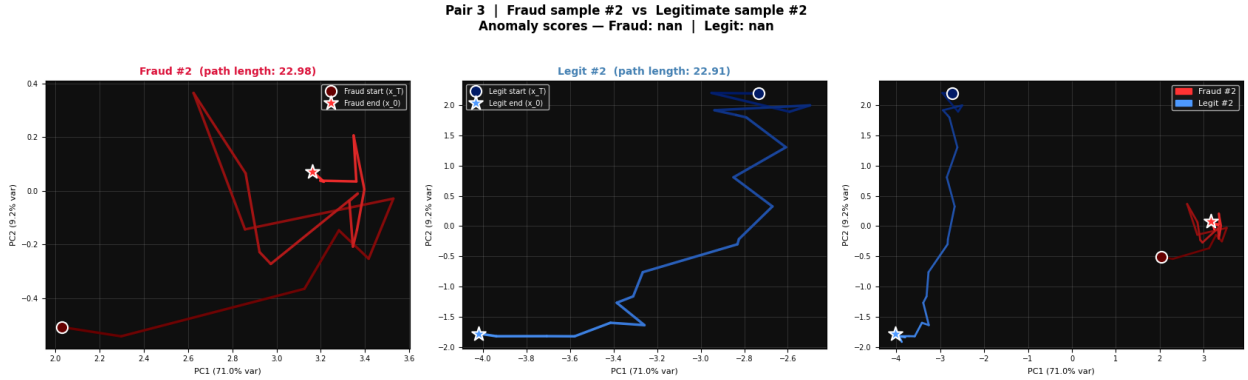


Figure 5: Individual trajectory Representation

4 Discussion

The results from Part I indicate that diffusion-based models are effective for tabular fraud detection when trained exclusively on legitimate data. In particular, the transition from a baseline MLP (Version A) to a residual architecture with timestep conditioning (Version B) leads to a substantial improvement in both AUROC and Average Precision. This suggests that architectural choices play a critical role in capturing the structure of the denoising process and, consequently, in improving anomaly detection performance. The improvement in Average Precision is particularly notable, as it reflects the model's ability to detect rare fraudulent transactions in a highly imbalanced setting. Although both models achieve relatively high AUROC, Version A's low Average Precision suggests that ranking performance alone is insufficient for effective fraud detection. In contrast, Version B shows more balanced performance across both metrics, indicating improved discrimination of minority class instances. The gains in Version B can be attributed to the introduction of residual connections and timestep embeddings. Residual connections enable the model to learn incremental corrections during denoising, while timestep embeddings provide explicit conditioning on the noise level, allowing better adaptation across diffusion steps. Together, these components improve noise prediction and, consequently, anomaly scoring. Despite these improvements, overlap in anomaly score distributions persists, indicating that perfect separation between legitimate and fraudulent transactions is not achieved. This reflects a limitation of reconstruction-based anomaly detection and highlights the importance of using evaluation metrics that account for class imbalance.

Through geometric analysis, we calculated five features for each of the 500 transaction samples: curvature, path length, directional consistency, step length, and straightness. We found that there was virtually no difference between these features for the two classes. This can be explained by the nature of the DDPM reverse process, which follows the same fixed mathematical formula for every transaction, regardless of its class. The geometry of the denoising path is not model-determined, but data-determined. But what differs between fraud and legitimate is not how the transaction moves through the feature space but how accurately the model reconstructs it at each time step. This was confirmed my the grid search, where it assigned zero weight to geometry. This narrows down on the future direction of the work, proving that geometric trajectory analysis is not a promising avenue. Instead, we should focus more on the reconstruction error itself.

Lastly, the interpretability analysis produced more valuable findings. We found that mid-range timesteps (41, 20, and 46) were the most discriminative. Additionally, we were able to narrow down the top 10 features (out of 30) that have the highest error gap between the two classes. PCA and T-SNE plots confirmed all our finding. PC1 alone captures 58.5% variance and aligned almost perfectly with the fraudulent/legitimate boundary without using any class labels. A tighter T-SNE cloud showed that the fraudulent cases are not only easily differentiated from the legitimate classes, but they are also similar to each other. The explanation generator translated all of this into actionable plain-language outputs for individual transactions, moving the system from a black-box model to something a fraud analyst can actually use and act accordingly.

5 Conclusion

In conclusion, this project shows that diffusion-based models are effective for unsupervised tabular fraud detection when trained exclusively on legitimate data. The transition from baseline MLP to a residual architecture with timestep encoding produced substantial improvements in both AUROC and average precision. This confirms that residual connections and timestep embeddings are critical architectural choices for improving anomaly scoring in highly imbalanced settings. Geometric trajectory analysis yielded a clean negative result. All five calculated geometric features were identical for the two classes. This confirmed that the geometry of the denoising path is model-determined and carries no discriminative signal. So, we moved on to incorporating explainability to project. It is important to note that the geometry was ***not used*** to drive the interpretability and explanation. Through this, we found that mid-range timesteps were the most discriminative, and we were also able to identify the top features with the highest error gap for the two classes. PCA and t-SNE were also able to confirm these findings. Together, these results move the system beyond the black-box model but towards something that is interpretable and can actually be used by analysts to drive change and identify fraud more easily.

6 Challenges

Initially, we had hypothesized that the geometry of the trajectory would look different for a fraudulent transaction and a legitimate transaction, and that these geometric differences could be used to drive the explainability. But as our results

have shown, that was not the case. So, we had to pivot slightly, and use only the reconstruction error (which works very well when it comes to differentiating the two classes) to derive human-interpretable explanations.

7 Member Contribution

Cynthia worked on Part I (i.e. evaluation of Tabular Diffusion for Fraud Detection) while Mahek worked on Part II (i.e. Analyses of the denoising trajectories and Explainability).

Acknowledgment

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